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Physics-Informed, Differentiable Hydrologic Models for Capturing Unseen Extreme Events

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Key Points:

- Differentiable models (DMs) better capture extreme events with magnitudes not included in the training data, compared to long short-term memory
- DMs optimized for extreme predictions can still offer good spatial generalization and robust predictions for untrained variables
- We theorize that DM's good extrapolation skill comes from physical constraints like mass conservation and storage-dependent flow

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Abstract

Recently, a hybrid framework combining machine learning (ML) and process-based equations, termed differentiable modeling, has shown comparable accuracy to pure ML models while offering enhanced interpretability and spatial generalizability. However, it remained unclear how well hybrid models generalize to extreme floods outside of the range of training data, and whether optimizing models for extreme events jeopardizes spatial generalizability and the physical significance of internal variables. Here we evaluated multiple versions of a differentiable model (δHBV1.0 and δHBV1.1p) for predicting unseen extreme events, and benchmarked them against a widely-applied long short-term memory (LSTM) network on the CAMELS data set. We found that both δHBV and LSTM models performed well, with δHBV1.1p outperforming LSTM for events with a return period of 5 years or more. This advantage was more pronounced as the return period increased (0.06 higher median Nash-Sutcliffe efficiency and lower peak flow errors for 80% of the 50-year or rarer events). Loss function choice had a larger impact on δHBV1.1p than on LSTM, and we showed the proper loss led to δHBV models that further surpassed LSTM in different performance aspects. Furthermore, allowing more dynamic parameters improved the extreme metrics, had no negative impact on spatial generalization, and exerted a minimal influence on the untrained variables. We hypothesize that δHBV's mass balance and first-order exchange terms help constrain and inform its responses to mitigate the underestimation of peaks compared to LSTM. We conclude that adopting interpretable structural priors can improve generalizability to unseen cases and thus increase model reliability to better inform stakeholder preparedness.

Plain Language Summary

Machine learning models have the potential to behave strangely in situations when they have rarely or never seen a type of event during training. Recently, differentiable hydrological models (DMs), which connect neural networks with physical models, have demonstrated a similar level of accuracy to state-of-the-art machine learning approaches (e.g., LSTM). However, it is not clear if DMs can more effectively capture extreme events unseen in the training data, which are considered the most dangerous. Further, if a DM is better suited to capture extreme events, we must ask if its outputs for untrained variables (variables for which the model isn't given training data) still make physical sense, since physical interpretability is a big promise of DMs. This work systematically answers these questions by comparing DMs with a well-known LSTM benchmark for predicting unseen extreme events. We found that LSTM tended to undershoot extreme events that fell outside the range of the training data. In comparison, DMs were better at predicting unseen flood magnitudes (i.e., had better extrapolation capabilities). We argue this is because they respect mass conservation and represent storage-dependent flow so that high inputs like precipitation, no matter how extreme, can be converted to accurate streamflow outputs.

1. Introduction

Flood forecasting is increasingly important due to the rising frequency and severity of flood events (Huong & Pathirana, 2013; Kundzewicz et al., 2014; Mahmoud & Gan, 2018). Accurate predictions of flood magnitude and timing are essential for mitigating impacts on communities, infrastructure, and ecosystems (Read & Vogel, 2016; Salman & Li, 2018). The development and application of machine learning techniques such as long short-term

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memory (LSTM) and Transformer networks have elevated flood forecasting to an unprecedented level of accuracy (Cai et al., 2024; Feng et al., 2020; Liu et al., 2024; Mosavi et al., 2018; Nearing et al., 2024; Nevo et al., 2021; Wi & Steinschneider, 2022). However, there remains a concern that pure machine learning (ML) models may lose reliability when forecasting extreme events, as such events are rarely or never seen in the training data sets. Additional concerns arise due to the lack of interpretability and physical constraints in pure ML models, but thus far ML models have outperformed traditional models on the prediction of extreme flow events (Frame et al., 2022).

Recently, differentiable modeling (DM) (Feng et al., 2022; Shen et al., 2023; Tsai et al., 2021) has been proposed as a genre of physics-informed machine learning that trains neural networks (NNs) and process-based models simultaneously in an "end-to-end" manner (Figure 1). "End-to-end" means all processes in a model between inputs and outputs are trained as a unified pipeline without the need for "ground truth" comparisons for intermediate outputs. This means every component of the model can be simultaneously updated based on the comparison of final outputs to observation data (most efficiently done using backpropagation). In a typical DM formulation, neural network modules provide parameters or missing processes, including time-dependent "dynamic" parameters, to the process-based equations (called priors), and one NN is trained and applied over the whole domain to maximally benefit from the commonalities of data (Fang et al., 2022; Tsai et al., 2021). Differentiability, supported by automatic differentiation on machine learning platforms like PyTorch, TensorFlow, JAX, and Julia, or adjoint operators (Rackauckas et al., 2021; Song, Knoben, et al., 2024), allows for tracking gradients through all components of the model and thereby enables efficient learning using gradient descent optimization methods. Automatic differentiation typically requires modelers to code only the forward model without the need to provide code for gradient calculation. DMs have achieved strong performance in streamflow simulation (Droppers et al., 2024; Feng et al., 2022, 2023; Jiang et al., 2020; Song, Knoben, et al., 2024; Wang & Gupta, 2024; Xie et al., 2021; Yang et al., 2023; Zhong et al., 2024), soil moisture simulation (Li et al., 2024), ecosystem modeling (Aboelyazeed et al., 2023; Reichstein et al., 2019), streamflow routing (Bindas et al., 2024), and stream temperature modeling (Rahmani et al., 2023), among other applications. In hydrological modeling, Feng et al. (2022) and Song, Shen, et al. (2024) have demonstrated that the performance of DMs approaches or equals that of state-of-the-art LSTM models, especially when an implicit numerical scheme is used. Additionally, DM shows advantages in spatial extrapolation across untrained regions (Feng et al., 2023; Li et al., 2025). A common concern about DMs is that their reliance on a process-based model structure may incur a performance penalty if the prior structure is not optimal for process representation (Rahmani et al., 2023) and may have slower development than purely data-driven models. Conversely, that same reliance may provide certain regularization benefits where relevant observational data is insufficient.

Notwithstanding the promising previous findings, further investigation is needed to determine whether a DM can accurately capture extreme events that are not shown in historical or training data. Previous evaluations have mostly focused on common metrics such as Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), and high flow bias (FHV), none of which are specific diagnostics of truly extreme flows that might occur only a handful of times in a given flow record. It is unclear if the relatively simple structures of process-based models could lead to higher errors in flood peak prediction compared to the pure ML models, or if the process-based structural priors could help to better infer unseen events. Also, DMs can have varied configurations, for example, depending on how many and which parameters are treated as being time-dependent. The objective of such dynamic parameters is to compensate for missing (but important) aspects of parameterizations such as vegetation and deep-water storage processes, and the impacts of these choices on modeling of extremes have not been adequately investigated. We have previously cautioned against freely applying extensive dynamic parameters due to the fear of model overfitting, which would increase the chance for equifinality and jeopardize interpretability (Feng et al., 2022; Song, Knoben, et al., 2024). Therefore, it remains unknown which model configurations are best suited to predict unseen events while maintaining interpretability.

Model configurations that better capture extreme events may impose a tradeoff between generalizability and the hybrid model's physical significance. In short, overfitting could degrade the model's spatial generalization ability. Pursuing optimal high flow representation may also degrade the prediction of internal variables or states due to the impacts of mass conservation, error compensation, and the "tug of war" between different processes. Due to model defects like numerical or structural errors, as well as data uncertainty, these trade-offs are often seen in simulations (Anache et al., 2019; Kavetski & Clark, 2010; Kollat et al., 2012; Song, Knoben, et al., 2024).

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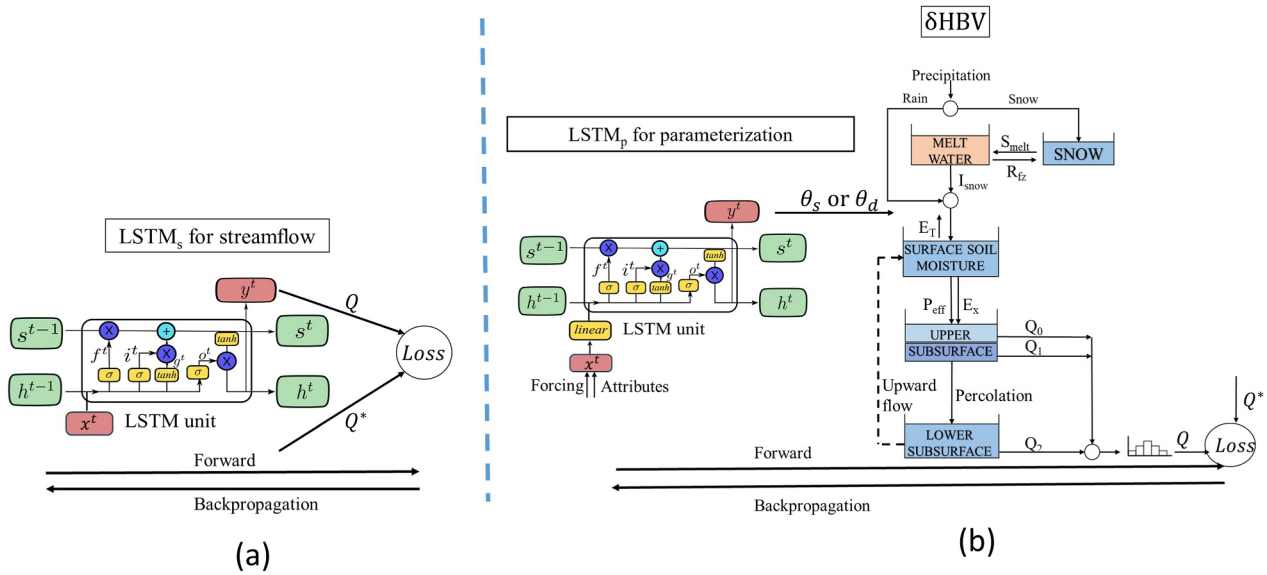

Figure 1. The schematic views of (a) LSTM$_s$ for streamflow and (b) δHBV. LSTM$_s$ in (a) is a long short-term memory (LSTM) model directly trained for streamflow prediction. LSTM$_p$ in (b) is trained for generating static and dynamic parameters for the Hydrologiska Byråns Vattenbalansavdelning (HBV) model, based on minimizing the loss between the streamflow simulation (Q) from the HBV model and the streamflow observation (Q^*). LSTM$_p$ provides temporally static (θ_s) or dynamic (θ_d) parameters for the HBV model. The upward flow (dashed arrow) in the HBV model is only used in δHBV1.1p but not in δHBV1.0. Additional explanations of the internal structures of LSTM and HBV modules can be found in Appendix B.

However, a counter-argument could be made that having more freedom in parameterization relaxes the structural constraints. The answer is not clear, and such issues thus need careful investigation.

In this work, we addressed the following questions using temporal and spatial tests for extreme events on a standard hydrological data set:

1. Can differentiable models capture extreme events when their training data sets do not include such magnitudes, and how do they compare to LSTM networks in this regard?
2. When models are better optimized to represent extremes, can they also provide good spatial generalization—that is, how effective are they in the PUB (Predictions in Ungauged Basins) test for extremes?
3. How robust are differentiable models optimized for extreme events at predicting other untrained variables, and does this change with different model configurations?

2. Data and Methods

2.1. Data Set

In this work, we used the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) data set for model benchmarking on extreme events (Addor et al., 2017; Newman & Clark, 2014). The CAMELS data set includes daily meteorological forcing data, static basin attributes, and streamflow observations from the United States Geological Survey (USGS) for 671 basins across the contiguous US. We used the same subset of 498 basins selected by Frame et al. (2022), and meteorological forcing data aggregated into basin-averaged values from Daily Surface Weather Data on a 1-km Grid for North America (Daymet) Version 4, as provided in CAMELS.

In addition to streamflow evaluation, we compared other untrained model outputs (meaning the model did not receive training data for these variables) to independent data sets. Snow water equivalent (SWE) was compared with daily 4-km gridded SWE data from the University of Arizona (UA) data set (Broxton et al., 2016), and evapotranspiration (ET) was compared with the 8-day 500-m MOD16A2 evapotranspiration/latent heat flux data set from 2001 to 2014 (Ladson et al., 2013). We aggregated the gridded products to the basin level using spatial means for comparison. For SWE, we selected 183 snow-dominated basins (daily mean SWE >10 mm from 1980 to 2020) from CAMELS for comparison. Here we used a basin-based (mean areal) evaluation to compare against

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various model versions to evaluate whether variables still carried physical significance when configurations were changed.

2.2. Models

We compared the performance of two types of models in predicting unseen extreme streamflow events. The first one was a pure deep learning streamflow model using the LSTM network (Figure 1a), and the second was a differentiable streamflow model (δHBV) that contained a neural network connected with a differentiable implementation of a conceptual hydrologic model, Hydrologiska Byråns Vattenbalansavdelning (HBV) (Figure 1b).

2.2.1. Long Short-Term Memory

Long short-term memory is a type of recurrent neural network that has recently been widely used in pure data-driven hydrological modeling, including applications in predicting streamflow, soil moisture, water quality, and snow (Fang et al., 2017; Feng et al., 2020; Kratzert et al., 2019; Ouyang et al., 2021; Rahmani et al., 2021; Song, Chaemchuen, et al., 2024; Song, Tsai, et al., 2024). It employs structures called memory cells (LSTM units) (Figure 1a), which include input, output, and forget gates, to selectively add, retain, or discard information over time (Graves, 2012; Hochreiter & Schmidhuber, 1997). These structures are specifically designed to address the vanishing gradient problem in deep learning (Hochreiter & Schmidhuber, 1997). In this work, we used the LSTM code from Kratzert et al. (2019) and trained it for streamflow prediction as a third-party reference for cross-group comparability (here referred to as $\mathrm{LSTM_s}$). $\mathrm{LSTM_s}$ can be simply viewed as a transformation from an input sequence to an output sequence:

$$Q^{1:t} = \mathrm{LSTM}_s\left(x_{\mathrm{s}}^{1:t}, A_{\mathrm{attr,s}}, W_{\mathrm{s}}\right) \tag{1}$$

where Q is the simulated streamflow. The meteorological forcings (x_{s}) vary with time (t) and contain Daymet forcing variables provided in CAMELS, including precipitation, solar radiation, maximum and minimum temperature, and vapor pressure. Attributes ($A_{\mathrm{attr,s}}$) include topography, climate, soil texture, land cover, and geology characteristics; the full attribute list can be found in Kratzert et al. (2019). W_{s} represents the weights and biases of the neural network. Additional LSTM structural details can be found in Appendix B1. Since $\mathrm{LSTM_s}$ is trained regionally (with a randomly-selected batch of basins trained together in each iteration), all time-dependent variables (such as x and Q) have three dimensions: time, batch (basins), and variables. In contrast, static variables ($A_{\mathrm{attr,s}}$) have two dimensions: batch and variables. Once the model is trained, it can be easily applied to the entire domain because the trained LSTM is regularized and shared across the domain.

2.2.2. Differentiable Streamflow Model

The differentiable streamflow model (δHBV) can be viewed as a regional HBV model with parameterization supported by a neural network (Figure 1b). δHBV first uses a neural network, which is also a LSTM network (called $\mathrm{LSTM_p}$), to output a list of physical parameters for the HBV models across a batch of basins:

$$\theta_{\mathrm{s}}, \theta_{\mathrm{d}}^{1:t} = \mathrm{LSTM_p}\left(x_{\mathrm{p}}^{1:t}, A_{\mathrm{attr,p}}, W_{\mathrm{p}}\right) \tag{2}$$

where x_{p} includes the time series of meteorological forcings—precipitation (P), mean temperature (T), and potential evapotranspiration (E_P, calculated with maximum and minimum temperature using the Hargreaves method (Hargreaves, 1994)). To be consistent with Frame et al. (2022) and Kratzert et al. (2019), more forcing variables were input to $\mathrm{LSTM_s}$ (Equation 1) than $\mathrm{LSTM_p}$ in the δHBV variants, which follow Feng et al. (2022). This might slightly benefit $\mathrm{LSTM_s}$. $A_{\mathrm{attr,p}}$ are the static attributes listed in Table A1 in Appendix A. W_{p} represents the weights and biases of the neural network. θ_{d} and θ_{s} represent the dynamic (time-variant) and static parameters of the HBV modules, respectively. θ_{s} includes all parameters in HBV except those selected to be in θ_{d}, as well as the routing parameters in the unit hydrograph routing formula.

δHBV uses HBV to simulate the conceptual hydrologic processes, including snow accumulation and melt, soil moisture, evapotranspiration, and runoff generation (Aghakouchak & Habib, 2010; Beck et al., 2020; Bergström, 1976, 1992; Seibert & Vis, 2012), using the parameters from $\mathrm{LSTM_p}$:

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$$Q^{1:t} = \mathrm{HBV}\left(x_{\mathrm{p}}^{1:t}, \theta_{\mathrm{s}}, \theta_{\mathrm{d}}^{1:t}\right) \tag{3}$$

where HBV represents all rainfall-runoff modules in the HBV model and Q is the simulated streamflow with units of mm/day.

The HBV structure used in this work is adapted from the HBV structure in Feng et al. (2022) (originally from Beck et al. (2020)) and incorporates an upward flux (C_r) proposed in Song, Knoben, et al. (2024). The model from Feng et al. (2022) is referred to as δHBV1.0 here. The new structure incorporating C_r is called δHBV1.1p. C_r was introduced to move water from the lower subsurface zone to the surface soil (as illustrated by the dashed arrow in Figure 1b), where the water can either be removed by evapotranspiration or contribute to flow later. The HBV equations and parameters, as well as the incorporation of C_r in δHBV1.1p, can be found in Table B1 in Appendix B. Here, we primarily focused on δHBV1.1p, using δHBV1.0 from Feng et al. (2022) solely for comparison. Thus, δHBV1.0 retained its original model parameter configuration and loss function from Feng et al. (2022), while we investigated the impacts of dynamic parameters and loss functions on δHBV1.1p, as detailed in the following sections.

2.2.3. Dynamic Parameterization

Instead of site-by-site parameter calibration, δHBV uses $\mathrm{LSTM_p}$ to directly learn a generalized (or regionalized) relationship between the inputs (meteorological forcings and geographic attribute data) and the physical parameters of HBV (Equation 2). This learning process is guided by an objective function defined to minimize the difference between the HBV streamflow simulations and observations, rather than being defined directly on the parameters output by $\mathrm{LSTM_p}$. Because of this direct linkage, no ground truth values for parameters are needed, although they could be incorporated as additional constraints if available. The model can track the gradients of the objective function with respect to the weights of $\mathrm{LSTM_p}$ through both the HBV and LSTM components, enabling the training of $\mathrm{LSTM_p}$. An additional benefit of this method is that $\mathrm{LSTM_p}$ can provide dynamic parameters that change daily, driven by the input meteorological forcing. Such dynamic parameters can compensate for unrepresented time-dependent processes in the original HBV model structure, such as vegetation growth and dynamics, as demonstrated in Feng et al. (2022). We can optionally select certain important parameters to be dynamic based on expert knowledge, while keeping the rest static to avoid overfitting. Feng et al. (2022) selected shape coefficients of effective flow (β) and evapotranspiration (γ) as dynamic parameters for δHBV1.0. In this work, we extended this by testing two additional dynamic parameters for δHBV1.1p that may influence extreme events: the recession coefficient of fast flow, θ_{K_0}, and the maximum upper subsurface (soil layer) storage, θ_{UZL}. θ_{K_0} and θ_{UZL} are the parameters that control the generation of near-surface flow (fast flow):

$$Q_0 = \theta_{K_0}(S_{\mathrm{UZ}} - \theta_{\mathrm{UZL}}) \tag{4}$$

where Q_0 is the fast flow from the upper soil layer and S_{UZ} is the upper soil layer storage.

Dynamic θ_{K_0} and θ_{UZL} can be related to the time-dependent connectivity and variable source areas in the watershed. In total, we tested three different dynamic parameterizations with δHBV1.1p: $\{\gamma, \beta\}$, $\{\gamma, \beta, \theta_{K_0}\}$, and $\{\gamma, \beta, \theta_{K_0}, \theta_{\mathrm{UZL}}\}$. All other parameters in HBV and routing equations are static. The definitions of all parameters are provided in Table B1 in Appendix B.

2.2.4. Optimization Configuration

δHBV1.0 from Feng et al. (2022) used a root-mean-square error (RMSE)-based loss function (Equation 5). With the same loss function, Song, Knoben et al. (2024) demonstrated that after incorporating C_r, δHBV1.1p improved the low flow prediction while maintaining its high flow performance.

$$\mathrm{Loss_{RMSE}} = (1.0 - \alpha)\sqrt{\frac{\sum_{b=1}^{B} \sum_{t=1}^{T} (\boldsymbol{Q}_b - \boldsymbol{Q}_b{}^*)^2}{B * T}} + \alpha\sqrt{\frac{\sum_{b=1}^{B} \sum_{t=1}^{T} \left(\widehat{\boldsymbol{Q}_b} - \widehat{\boldsymbol{Q}_b{}^*}\right)^2}{B * T}} \tag{5}$$

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Q_b and $Q_b{}^*$ are respectively the streamflow simulation and observation of basin b, while the log-transformed versions are $\widehat{Q_b} = log_{10}(\sqrt{Q_b} + 0.1)$ and $\widehat{Q_b^*} = log_{10}(\sqrt{Q_b^*} + 0.1)$. α is a weight to balance the model's performance between high flow and low flow (0.25 as selected by Feng et al. (2022)), B is the number of basins in a batch, and T is the time series length of each batch.

In this work, we tested two additional loss functions for both $LSTM_s$ and δHBV1.1p, based on mean-normalized-square error (MNSE, Equation 6) and KGE (Equation 7). In the MNSE-based loss function, peak flows with significant deviations have proportionally larger impacts on the loss. It has been widely used in the training of LSTM streamflow models (Frame et al., 2022; Kratzert et al., 2019).

$$\mathrm{Loss}_{\mathrm{MNSE}} = \frac{1}{B}\sum_{b=1}^{B}\frac{\sum_{t=1}^{T}\left(\boldsymbol{Q}_b^t - \boldsymbol{Q}_b^{t*}\right)^2}{\left(\mathrm{std}\left(\boldsymbol{Q}_b^*\right) + \epsilon\right)^2} \tag{6}$$

Here, $\mathrm{std}\left(Q_b^*\right)$ is the standard deviation of the observed runoff in basin b, which avoids overweighting large and/or wet basins in the loss function and ϵ is a small value (0.1) to stabilize the gradient calculation.

The KGE-based loss function has demonstrated better performance in predicting high-flow events (Mizukami et al., 2019) than square-error-based loss functions such as Equations 5 and 6. The KGE-based loss function is similar to 1-KGE:

$$\mathrm{Loss}_{\mathrm{KGE}} = \sqrt{(a-1)^2 - (b-1)^2 + (r-1)^2} \tag{7}$$

where a is the Pearson's correlation coefficient between all corresponding streamflow simulations and observations for the basins in the batch; b and r are the bias and variability ratios, respectively.

All three loss functions were applied to the simulations of a randomly selected batch of basins in one training iteration, a common strategy in large-scale ML models to improve generalization and prevent overfitting. Although these loss functions do not specifically represent the extreme flow performance, they effectively evaluate model skill across all flow magnitudes. By contrast, extreme flow metrics (e.g., annual peak flow bias) produce the best peak performance at the expense of overall fidelity (Mizukami et al., 2019), so we did not use them as loss functions here. There are also many other loss functions, such as the hydrograph functionals (e.g., flow duration curve divergence scores) proposed by Vrugt (2024). However, the loss functions for differentiable models need to be programmatically differentiable (e.g., they cannot involve integration, sorting, or indexing functions) and efficient for large-scale training. Thus, some loss functions that are applicable to traditional hydrologic models for single-basin simulations—potentially making the simulations more hydrologically realistic—are not suitable for differentiable models (DMs). Here, we primarily focused on the comparison between the purely data-driven LSTM and hybrid differentiable models in unseen extreme events. Further research is needed on the selection of appropriate loss functions.

2.3. Model Configurations and Hyperparameters

To avoid overtuning, we directly used model hyperparameters from the literature, as $LSTM_s$ is from Frame et al. (2022) and $LSTM_p$ in δHBV is from Feng et al. (2022). All models used a batch size of 100 and a time series length of 365 days. $LSTM_s$ used a hidden size of 128 while $LSTM_p$ used a hidden size of 256 with a dropout factor of 0.5. Thus, both LSTMs have the same effective number of hidden cells. δHBV variants were trained using the AdaDelta optimizer to automatically adjust the learning rate. $LSTM_s$ used the Adam optimizer with a fixed learning rate schedule, starting at 0.001 and decreasing to 5×10^4 after 10 epochs and 1×10^4 after 20 epochs. While the purely data-driven $LSTM_s$ for streamflow prediction and $LSTM_p$ for δHBV parameterization serve different purposes, they have very similar structures (detailed in Appendix B).

2.4. Training and Testing Specifications

For each basin in the CAMELS streamflow data set, we used annual peak flow observations from the USGS National Water Information System to calculate the return period for each water year, following the guidelines in the U.S. Interagency Committee on Water Data Bulletin 17b, which fits log-transformed annual peak flows of

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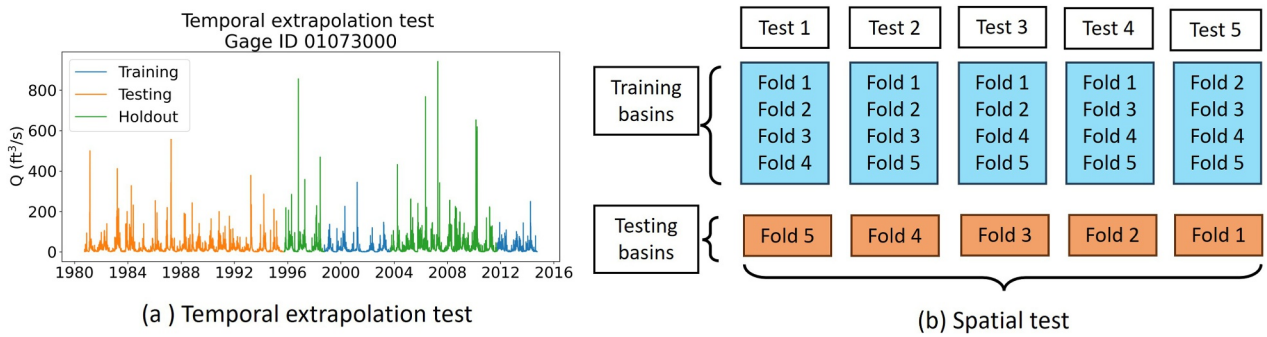

Figure 2. Overview of the two experiments to evaluate the extrapolation ability of the model in the prediction of unseen extreme events. (a) Temporal extrapolation test, where "holdout" (green) refers to the data excluded from training due to that year having a greater than 5-year return period peak flow. (b) 5-fold spatial test, where each model is trained and tested on different folds of basins for a total of five rounds. Test results from all basins are collected to calculate performance metrics.

each basin to a Pearson type III distribution (Frame et al., 2022; Interagency Advisory Committee on Water Data, 1981). We designed two experiments to compare $LSTM_s$ and the δHBV variants for predicting unseen extreme events: a temporal extrapolation test and a spatial test (Figure 2).

In the temporal extrapolation test, we trained all models on water years with a maximum flow return period of 5 years or lower from 1995/10/01 to 2014/09/30, holding out water years with a greater than 5-year return period peak flow within this time period for each basin. We tested all models from 1980/10/01 to 1995/09/30 (Figure 2a). Instead of directly testing the models on the holdout water years from 1995 to 2014, we used this separate time period to better assess the models' temporal extrapolation capabilities.

In the spatial test (similar to the common Prediction in Ungaged Basins, or PUB, test), we similarly used water years with a 5-year or lower return period peak flow from 1990/10/01 to 2014/09/30 for training and held out the water years with greater than a 5-year return period peak flow. The training time span was selected to be consistent with Frame et al. (2022). The spatial test was conducted using a 5-fold cross-validation approach that randomly divided all 498 basins into 5 folds. We trained the model on 4 folds of basins with >5-year return period water years held out, and tested the model on the remaining fold of basins for all water years from 1990/10/01 to 2014/09/30 (Figure 2b). The spatial test on untrained basins is to show the models' spatial generalizability. We ran the test 5 times, rotating the testing fold each time so that all basins served as a test basin once (orange boxes in Figure 2b); results are thus reported for all basins.

For performance metrics, we looked at absolute fraction errors of annual peak flow, the NSE (Nash & Sutcliffe, 1970), the KGE (Gupta et al., 2009), the root-mean-square errors of low flows (the bottom 30% of the flow range), high flows (top 2% of the flow range), and middle flows (flow magnitude between bottom and peak flows), and the absolute percent bias of peak flows. Their definitions and equations are provided in Table 1.

3. Results

This section presents the results of the experiments designed and conducted in this study, while the following Discussion section compares our models with those in the existing literature and offers reflections on the differentiable models. Results from benchmarking experiments, commonly used in studies of LSTM and hybrid models, are included in Appendix C for interested readers.

3.1. LSTMs vs δHBV Variants

In the temporal extrapolation test, the differentiable models (δHBVs) generally presented smaller peak errors than $LSTM_s$ in low-probability years (years with return periods of annual peak flow >5 years in Figure 3a, Table 2). The performance gaps between $LSTM_s$ and all δHBV models are small for events with return periods less than 50 years, while the gaps become noticeably larger for very rare events (return periods >50 years), which are more dangerous and of greater interest for real-world applications. For extreme events with return periods greater than 50 or 100 years, δHBV1.1p reduced the error for 80% of these events compared to $LSTM_s$, and the difference in peak error was larger than 10% (absolute value) in 40% of events and larger than 20% (absolute value) in 12% of events (Figure 4a). These observations support the hypothesis that differentiable models with process-based

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Table 1
Metrics Used to Evaluate Model Performance

Metrics	Definitions	Formulas
Absolute fraction error of annual peak flow (ARTB)	The absolute value of the relative bias between the observed annual peak flow and the simulated annual peak flow.	$ARTB = \left\lvert \frac{Q^{\text{sim}}_{\text{max,annual}} - Q^{\text{obs}}_{\text{max,annual}}}{Q^{\text{obs}}_{\text{max,annual}}} \right\rvert$ $Q^{\text{sim}}_{\text{max,annual}}$ and $Q^{\text{obs}}_{\text{max,annual}}$ are the simulated and observed annual peak flows, respectively.
Nash-Sutcliffe Efficiency (NSE)	The ratio of the model error variance to the observed variance, where 1 indicates a perfect model, and 0 indicates performance equivalent to predicting with the long-term mean.	$NSE = 1 - \frac{\sum_{i=1}^{N}\left(Q_i^{\text{obs}} - Q_i^{\text{sim}}\right)^2}{\sum_{i=1}^{N}\left(Q_i^{\text{obs}} - \overline{Q}_i^{\text{obs}}\right)^2}$ N is the number of days used in the evaluation. Q_i^{obs} and Q_i^{sim} represent the streamflow simulation and observation, respectively. $\overline{Q}_i^{\text{obs}}$ is the mean value of the observation.
Kling-Gupta Efficiency (KGE)	A comprehensive assessment of Pearson's correlation coefficient (a), bias ratio (b), and variability ratio (r).	$KGE = 1 - \sqrt{(a-1)^2 - (b-1)^2 + (r-1)^2}$
Root-mean-square error (RMSE) of flow	RMSE between streamflow simulation and observation.	$RMSE = \sqrt{\frac{\sum_{i=1}^{N}\left(Q_i^{\text{sim}} - Q_i^{\text{obs}}\right)^2}{N}}$
|FHV|of 0.1% flow	The absolute value of the percent bias of top 0.1% flows.	$\lvert FHV \rvert = \frac{\left\lvert \sum_{i=1}^{N} Q_i^{\text{sim}} - \sum_{i=1}^{N} Q_i^{\text{obs}} \right\rvert}{\sum_{i=1}^{N} Q_i^{\text{obs}}}$

modules as their backbones are more reliable in extrapolation than pure machine learning models. The NSE and KGE comparisons illustrate similar trends (Table 2 and Figure C1 in Appendix C1).

Compared with δHBV1.1p trained with the MNSE loss function, using the KGE-based loss function further reduced the peak errors of δHBV1.1p by around 5% for each flow category, presenting a larger improvement when compared to $LSTM_s$ trained with the same loss (Figure 3c). In contrast, δHBV1.0 with the RMSE loss function had the lowest performance among all δHBVs but still outperformed $LSTM_s$ (even the one trained with KGE-based loss) for return periods >50 years. Using KGE loss also improved the unseen-event performance for $LSTM_s$, but the effect was less pronounced. However, when the training data were complete (no holdout), δHBV1.1p trained with the KGE loss showed a substantial performance drop compared with the one using the MNSE loss (see Appendix C2; more discussion can be found in the following section). Thus, to maintain overall model fidelity and for comparison with the literature that uses MNSE loss (Frame et al., 2022; Kratzert et al., 2019), we produce the subsequent results of $LSTM_s$ and δHBV1.1p with MNSE loss.

In the spatial test, we observed a similar trend: $LSTM_s$ and δHBVs achieved similar performance for return periods <50 years, while δHBVs showed a clear advantage in simulating infrequent events (return periods >50 years) across annual peak flow error (Figures 3b and Table 2) and NSE and KGE scores (Figure C1, Appendix C). δHBVs reduced the median peak errors by more than 7% in very low-probability events (>100-year events in Figure 3b). Specifically, δHBV1.1p reduced the peak errors for 75% of rare extreme events (return periods >50 or 100 years) compared to $LSTM_s$ (Figure 4b) and reduced peak errors by over 10% (absolute value) for around 35% of rare events and by over 20% (absolute value) for around 20% of rare events.

By examining flow biases and hydrographs in detail, we see that δHBV1.1p accurately predicted the large magnitudes of the unseen extreme events that $LSTM_s$ underestimated by 30%–40% (Figures 5a and 5c). As the largest discharge values of the training data were ∼3 mm/day (Figure 5a) and 35 mm/day (Figure 5c), when confronted with large storm inputs, $LSTM_s$'s cell states could become saturated, causing its failure to predict a proportionally larger response. Meanwhile, δHBVs (driven by mass conservation) were able to capture the almost linear rainfall-runoff relationship during extreme floods. Furthermore, due to the increase in groundwater storage alongside the peak runoff, δHBVs accurately predicted higher baseflow in the post-peak recession, whereas LSTMs tended to underestimate it (Figure 5a). Additionally, by explicitly accounting for the long-term water storage dynamics, δHBVs also better captured the large peaks associated with snow accumulation and melt processes (Figure 5c), where the streamflow peak significantly exceeded the precipitation peak due to the release of stored water in the snowpack— a process ignored by $LSTM_s$. While these events are inherently rare, they are

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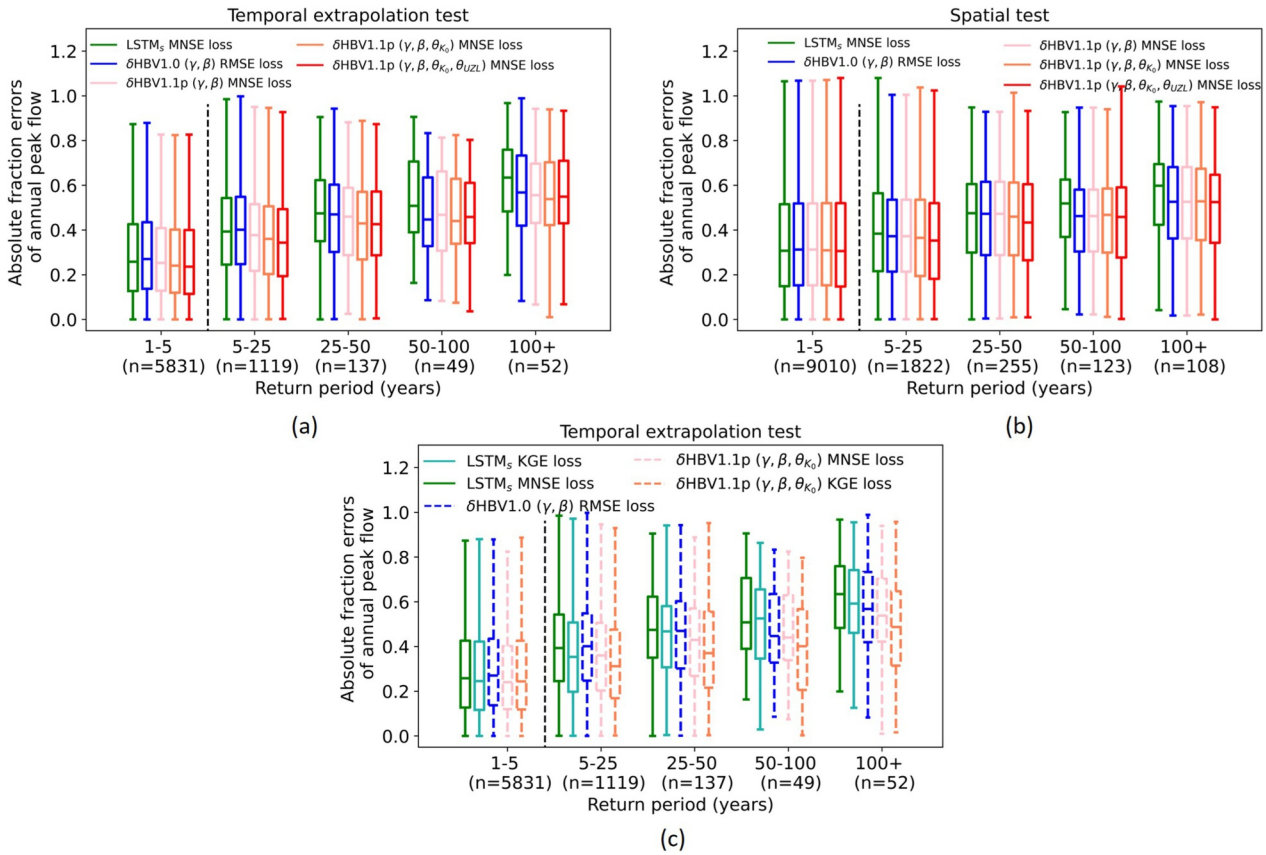

Figure 3. Box plots of the absolute fraction errors of annual peak flow stratified by the peak flow return periods of the water years in the (a) temporal extrapolation test and (b) spatial test on untrained basins, while (c) shows a comparison of models using different loss functions with temporal test results. The *n* in the *x*-axis label represents the total number of basin-years collected in each category. LSTM$_s$ is the pure data-driven long short-term memory network for streamflow prediction. δHBV1.0(γ,β) is the differentiable streamflow model from Feng et al. (2022) and δHBV1.1p(γ,β), δHBV1.1p($\gamma,\beta,\theta_{K_0}$), and δHBV1.1p($\gamma,\beta,\theta_{K_0},\theta_{\mathrm{UZL}}$) are the differentiable streamflow model variants developed in this work using different dynamic parameters. MNSE loss indicates the loss function was based on mean-normalized-square error, RMSE loss was based on root-mean-square error, and KGE loss was based on Kling-Gupta Efficiency. Each box's horizontal line represents the median, while its top and bottom mark the first and third quartiles. The whiskers extend to the lowest and highest points within 1.5 times the interquartile range.

Table 2
Summary of Median Annual Streamflow Simulation Metrics in the Temporal Extrapolation and Spatial Tests

	Temporal extrapolation test in low-probability years (return period >5 years)					Spatial test in low-probability years (return period >5 years)				
Model	NSE	KGE	Low flow RMSE (mm/day)	Middle flow RMSE (mm/day)	High flow RMSE (mm/day)	NSE	KGE	Low flow RMSE (mm/day)	Middle flow RMSE (mm/day)	High flow RMSE (mm/day)
LSTM$_s$ MNSE loss	0.734	0.613	**0.065**	0.491	7.511	0.692	0.609	0.106	0.586	8.005
δHBV1.0 (γ,β) RMSE loss	0.724	0.617	0.076	0.496	7.402	0.688	0.615	0.108	0.597	8.093
δHBV1.1p (γ,β) MNSE loss	0.753	0.656	0.069	0.442	6.971	**0.697**	0.623	0.107	0.575	7.937
δHBV1.1p ($\gamma,\beta,\theta_{K_0}$) MNSE loss	**0.759**	**0.663**	0.068	**0.441**	6.931	0.694	0.624	0.107	0.582	7.680
δHBV1.1p ($\gamma,\beta,\theta_{K_0},\theta_{\mathrm{UZL}}$) MNSE loss	**0.759**	**0.663**	0.069	0.452	**6.759**	**0.697**	**0.633**	**0.102**	**0.569**	**7.480**

Note. The parameters denoted in parentheses are the parameters set to be dynamic for each model. Bold values denote the best metric in each column.

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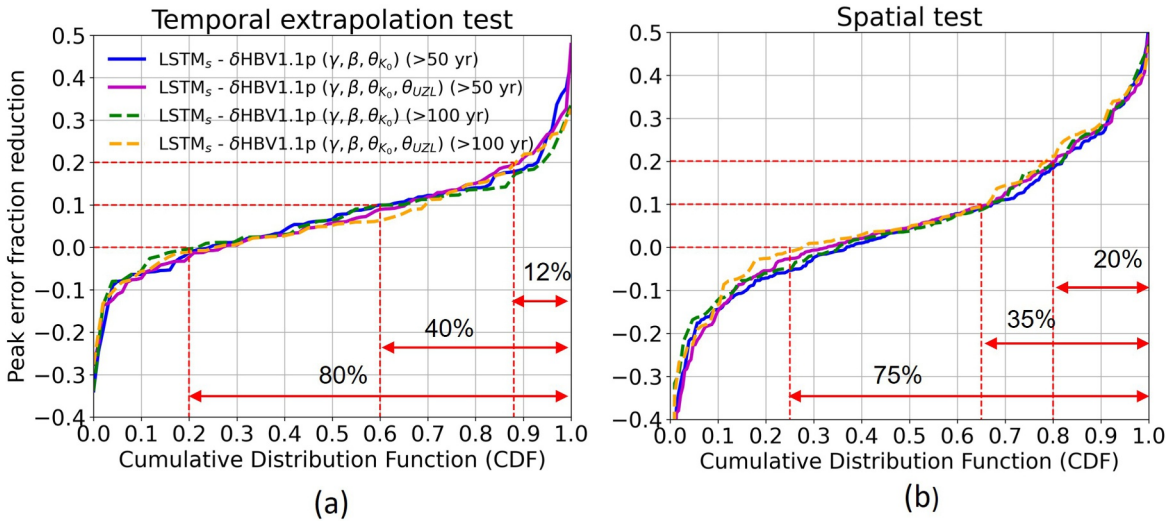

Figure 4. Cumulative Distribution Functions of peak error fraction reduction by differentiable models (using LSTM$_s$ as a baseline to calculate the difference of the absolute fraction peak errors) in low-probability basin-years. (a) Temporal extrapolation test and (b) spatial test results for models using the MNSE loss. Solid lines indicate results for events with return periods over 50 years, while dashed lines indicate return periods over 100 years. The straight red dashed lines indicate the percentages of low-probability basin-years where δHBV1.1p (left to right) outperforms LSTM$_s$, reduces peak errors by 10%, and reduces peak errors by 20%, respectively.

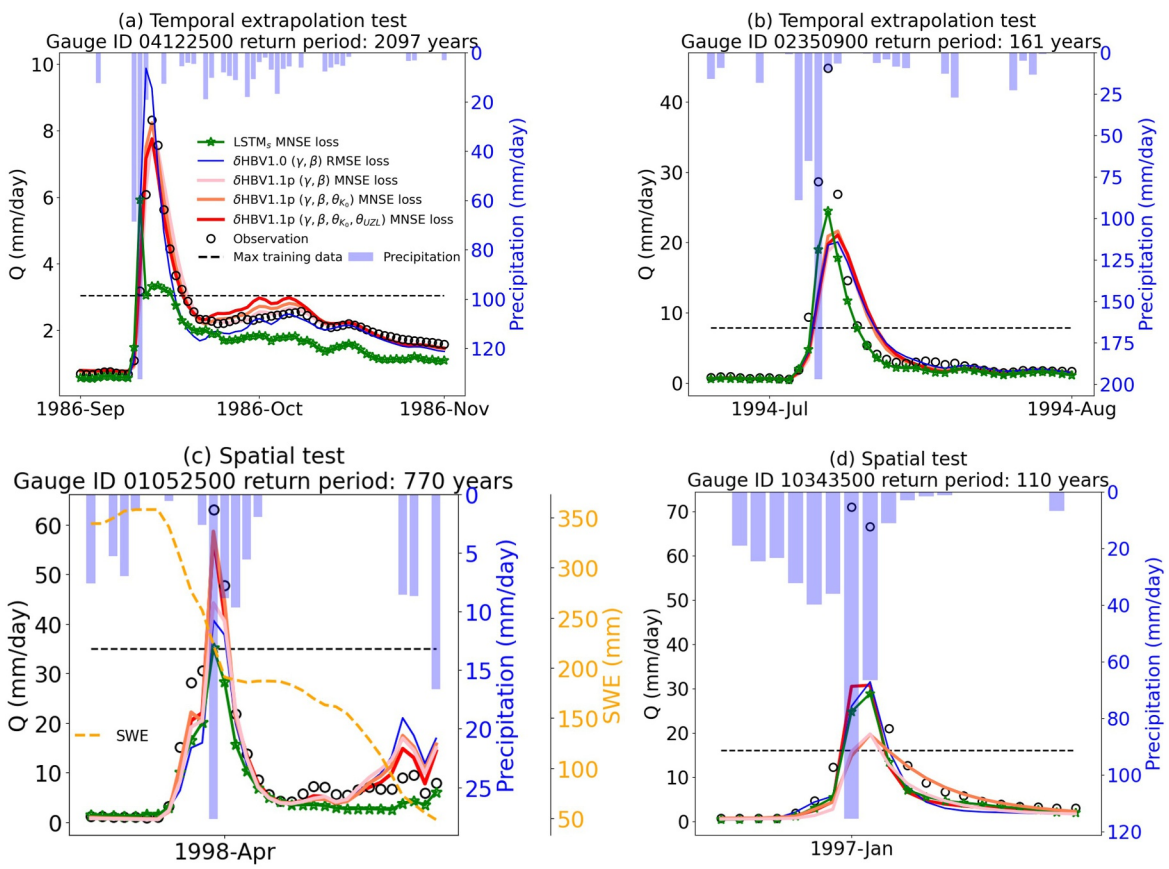

Figure 5. Hydrographs of all models in the temporal extrapolation test at (a) Pere Marquette River at Scottville, MI–04122500 and (b) Kinchafoonee Creek at Pinewood Road, near Dawson, GA–02350900, and in the spatial test at (c) Diamond River near Wentworth Location, NH - 01052500 (δHBV-simulated Snow Water Equivalent is plotted as a tertiary *y*-axis) and (d) Sagehen Creek near Truckee, CA–10343500. Sites (a, b) were randomly selected from extreme events where δHBVs showed advantages, while sites (c, d) were selected from extreme events where LSTM$_s$ performed better. The horizontal dashed black line indicates the maximum streamflow observation from the training data.

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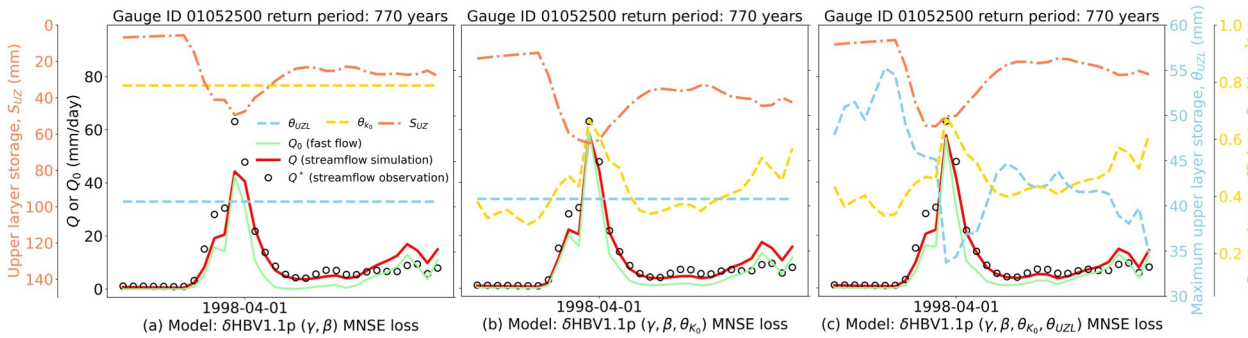

Figure 6. Plots of streamflow (Q), simulated fast flow (Q_0), water storage in the upper soil layer (S_{UZ}), and two estimated parameters (θ_{K_0} and θ_{UZL}) from δHBV1.1p variants with different numbers of dynamic parameters: (a) δHBV1.1p (γ,β), (b) δHBV1.1p ($\gamma,\beta,\theta_{K_0}$), and (c) δHBV1.1p ($\gamma,\beta,\theta_{K_0},\theta_{UZL}$). Each plot is for the gauge at Diamond River near Wentworth Location, NH–01052500 during the extreme event depicted in Figure 5c. The left y-axis (black text) corresponds with streamflow simulation (red lines), observation (black circles), and simulated fast flow (green lines). The secondary inverted left y-axis (orange text) corresponds with the upper soil layer storage (orange lines). The right y-axis (blue text) corresponds with θ_{UZL} (blue lines). The secondary right y-axis (yellow text) corresponds with θ_{K_0} (yellow lines).

exactly the events that can cause the biggest damage, and are when flood forecasting accuracy is the most important.

There were still some peaks where $\mathrm{LSTM_s}$ did better than δHBVs (Figures 5b and 5d), but in these cases, often both δHBVs and $\mathrm{LSTM_s}$ performed poorly, and the differences between them were not large. Such errors cannot be attributed to model structural errors (as both data-driven and process-based models failed) but rather to errors in the streamflow data, to errors in the gridded forcing data, or to errors introduced by basin averaging the forcings. $\mathrm{LSTM_s}$ may internally learn and mitigate some systematic errors in meteorological forcings or observations, allowing for slight corrections in peak timing (Figure 5b) or magnitude (Figures 5b and 5d) since $\mathrm{LSTM_s}$ has demonstrated its ability to leverage multi-forcing data sets to improve streamflow prediction (increasing median NSE from 0.74 to 0.80 for the CAMELS data set) (Kratzert et al., 2021). In contrast, due to the model structure of HBV, δHBV can only directly utilize a single forcing data set and requires additional processing steps, such as using a neural network for forcing fusion or corrections (Sawadekar et al., 2025). Despite this limitation, the advantage of δHBV over $\mathrm{LSTM_s}$ is still more widespread and significant, occurring in approximately 80% of extreme events (return period >50 years), with improvements exceeding 10% in half of these events, as discussed earlier (Figure 4).

3.2. Effects of Dynamic Parameterization in δHBV1.1p

Augmenting δHBV1.1p with two more dynamic parameters (θ_{K_0} and θ_{UZL}) generally helped elevate model performance, rather than causing overfitting. δHBV1.1p with three ($\gamma,\beta,\theta_{K_0}$) or four ($\gamma,\beta,\theta_{K_0},\theta_{\mathrm{UZL}}$) dynamic parameters both showed improvements compared to δHBV1.1p with two (γ,β) dynamic parameters (Figure 3). Feng et al. (2022) used the dynamic shape coefficient (β) and evapotranspiration coefficient (γ) in δHBV1.0 to represent the nonlinear relationship between surface soil moisture and effective rainfall, as well as nonlinear evapotranspiration. In this work, we introduced two more optionally dynamic parameters: the recession coefficient of fast flow from the upper soil layer (θ_{K_0}) and the maximum storage of the upper soil layer (θ_{UZL}). The dynamic θ_{K_0} reflects the dynamic water release rate due to changing groundwater levels, bank and wetland storages, and other factors. The dynamic θ_{UZL} represents storage capacity changes due to seasonally varying saturation excess, soil freezing and thawing, soil swelling, surface sealing in dryland soils, hysteretic water storage, and other factors.

Both θ_{UZL} and θ_{K_0} can influence fast flows (Equation 4) from the upper layer, and thus also influence peak flows. When both are static, as in δHBV1.1p(γ,β), a large θ_{K_0} is estimated overall to accommodate the increased fast flow during peak events (Figure 6a). The upper layer storage remains small overall, so there is little or no fast flow during low flows. Consequently, the model underestimates peak flow in an extreme event due to relatively low upper layer storage (Figure 6a). When θ_{K_0} is dynamic, such conflicts can be resolved. It remains small during low flow periods and increases during peak days. This dynamic behavior allows the model to maintain a larger upper layer storage by producing a small baseflow with a smaller θ_{K_0}, while increasing θ_{K_0} during large storms to

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Table 3
Summary of Median Snow Water Equivalent and Evapotranspiration Metrics

Model	Temporal test (SWE:1980/10/01–1995/09/30)		K-fold spatial test (SWE: 1990/10/01–2014/09/30, E_T : 2001/01/01–2014/09/30)		
	SWE NSE	SWE annual relative peak errors	SWE NSE	SWE annual relative peak errors	E_T NSE
δHBV1.0 (γ,β) RMSE loss	**0.794**	27.7%	**0.798**	**25.2%**	0.602
δHBV1.1p (γ,β) MNSE loss	0.784	**27.4%**	0.794	25.8%	0.683
δHBV1.1p $(\gamma,\beta,\theta_{K_0})$ MNSE loss	0.785	27.8%	0.786	26.0%	**0.700**
δHBV1.1p $(\gamma,\beta,\theta_{K_0},\theta_{\mathrm{UZL}})$ MNSE loss	0.771	29.1%	0.773	26.7%	0.680

Note. We only compared SWE at 183 snow-dominated basins (daily mean SWE >10 mm in 1980–2020) in the CAMELS data set. E_T is compared across all 498 basins from 2001 to 2014 when the satellite data was available. We caution that these metrics were obtained with basin-averaged forcings for comparison between model versions. Bold values denote the best value in each column.

produce a large peak (Figure 6b). When θ_{UZL} is dynamic, it can shrink water storage capacity before extremes to increase the runoff magnitudes (Figure 6c), potentially capturing the effects of soil surface sealing (in arid regions) or variable source areas (in humid regions). The amplification of fast flow caused by dynamic θ_{UZL} and θ_{K_0} can accumulate until it reaches the total water storage in the upper soil layer. For example, since dynamic θ_{K_0} already induces a fast flow with magnitude approaching S_{UZ} (Figure 6b), introducing an additional dynamic parameter, θ_{UZL}, cannot further increase the fast flow (Figure 6c). However, when the water storage is sufficient, using both dynamic θ_{UZL} and θ_{K_0} can help further reduce the underestimation of peaks. Due to this limitation, the advantage of δHBV1.1p with four parameters $(\gamma,\beta,\theta_{K_0},\theta_{\mathrm{UZL}})$ over three parameters $(\gamma,\beta,\theta_{K_0})$ is marginal. Overall, the use of dynamic parameters prevents the model from having to compromise between different climate, storage, or soil conditions due to unrepresented processes.

3.3. Evaluation of Untrained Variable Predictions

Another advantage of δHBV is its ability to provide valuable predictions of untrained intermediate variables within its process-based modules. The accuracy of these intermediate variables is used to demonstrate that the model correctly captures rainfall-runoff interactions between processes, rather than merely using the neural network to compensate for high streamflow performance. All the δHBV model variants provided reasonable internal states like SWE, and flux variables like evapotranspiration (E_T), in both temporal and spatial tests (Table 3). The NSE value for SWE prediction reached 0.77, and the annual relative peak error of SWE was less than 30%. The NSE value for E_T prediction reached 0.7 with δHBV1.1p by incorporating an upward flux, which is active only when soil moisture is low (see soil moisture and evapotranspiration module in HBV, Appendix B2). It can decrease low flow RMSE by removing extra water in the lower subsurface zone (Figure 1b) on dry days for the predictions of zero or near-zero flows (Song, Knoben, et al., 2024).

With more dynamic parameters, the performance of δHBVs in SWE and E_T prediction remained stable, demonstrating that the physical parameters derived from neural networks and the internal processes were not significantly distorted by a single target variable during training and were well-constrained by the physical HBV equations. For the same extreme event shown in Figure 6, all δHBV1.1p models with different dynamic parameters provided similar E_T and SWE simulations (Figure 7). The predicted SWE peak magnitudes and trends are comparable; however, allowing more dynamic parameters results in increased snowmelt with additional water to the soil layer. This internal adjustment helps alleviate the underestimation of extreme events, while the additional parameter dynamics can also prevent the overestimation of regular flows when more meltwater is received (see the post-peak recession in Figures 6b, 6c, and 7b).

4. Discussion

The good performance of δHBVs in the temporal extrapolation test showcases their generalization and extrapolation capabilities for unseen, rare flow events (Figure 3). Even though $\mathrm{LSTM_s}$ has demonstrated substantially higher performance than traditional hydrological models (Feng et al., 2020; Kratzert et al., 2019; Liu et al., 2024), making it challenging to surpass, the improvements achieved by δHBVs over LSTMs are still widespread. To benchmark against the literature and the reported traditional model, we tested our δHBV variants using the

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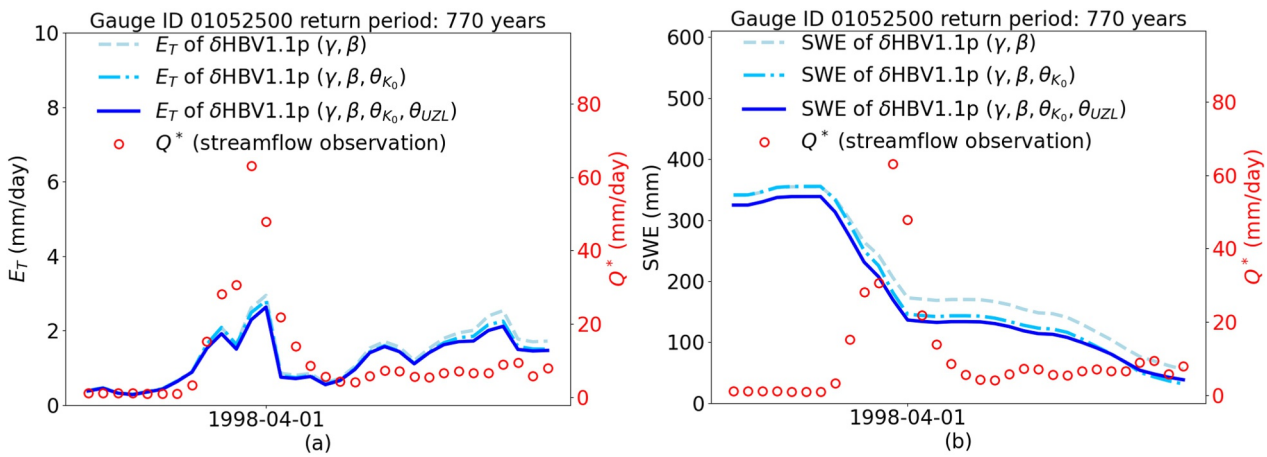

Figure 7. Plots of simulated (a) evapotranspiration (E_T) and (b) snow water equivalent from δHBV1.1p variants with different numbers of dynamic parameters for the gauge at Diamond River near Wentworth Location, NH–01052500 during the extreme event depicted in Figures 5c and 6. The right-hand *y*-axis (red text) corresponds with the streamflow observation.

experiment from Frame et al. (2022) (Appendix C3). This experiment, referred to as the temporal interpolation test (Figure C3a), predicts the holdout data (water years with return periods >5 years) for a timespan in between two training time periods rather than predicting for a separate time span as done in the temporal extrapolation test (Figure 2a). The performance gap between $LSTM_s$ and δHBVs narrowed in the temporal interpolation test (Figure C3b) compared to the temporal extrapolation test (Figure 3a). For example, $LSTM_s$ could outperform δHBV1.1p when the return period was <25 years. However, the interpolation test makes it easier to capture nonstationarity. Due to climate and catchment nonstationarities, there are new patterns and drifts (physical changes in the catchments) that were not present in historical data series. Training in two timespans with a gap in the middle for testing puts a bound on the nonstationarities, or, stated in a different way, the model can interpolate between system behaviors. Additionally, some basins can have interannual memory due to deep water storage that could impact streamflow responses, which is easier to capture in the interpolation test. The temporal extrapolation test designed in this work is more challenging, as the absence of training data over a long time period leads to a more open-ended and more relevant prediction problem. Moreover, pure machine learning models exhibit some benefits in inferring peak flows that share similar patterns with the training data by memorizing training patterns and noise.

The overall results show that hybrid δHBV models are better at "extrapolation" (untrained spatial locations or flow magnitudes), while $LSTM_s$ is strong at "interpolation" (trained spatial locations or flow magnitudes). Comparing the structures of $LSTM_s$ and δHBV (Figure 1), the superior extrapolation capability of δHBV over $LSTM_s$ mainly comes from δHBV's process-based modules. The mass balance calculations and assumed first-order exchanges between water storages impose structural rigidity — that is, if the system receives an unprecedented amount of rainfall, the water will be directed through the storage compartments and toward the outlets based on the specified hydrologic processes while respecting conservation of mass, leading to unprecedented streamflow. The first-order flux calculations also ensure that a higher amount of water storage in the groundwater module (see the runoff generation module in Appendix B2) of δHBV necessarily leads to a higher baseflow rate coming from it (Figure 5a). In contrast, $LSTM_s$, as a deep neural network, can behave erratically when extrapolating beyond the bounds of the training data (Guo et al., 2017; Nguyen et al., 2015). Additionally, the bounded nonlinear activation functions (such as sigmoid and hyperbolic tangent functions) used across various layers of $LSTM_s$ (Appendix B1) limit its ability to make predictions beyond its training range, often leading to under-predictions of peak flows (Figures 5a and 5c) (Baste et al., 2025).

Both $LSTM_s$ and δHBV models showed significant performance improvements over the locally-calibrated Sacramento Soil Moisture Accounting (SAC-SMA) model in the temporal interpolation test (Figure C3b of Appendix C3). Although this is not a completely fair comparison, given that HBV and SAC-SMA have different model structures, it nonetheless highlights the power of δHBV's NN ($LSTM_p$) for large-scale parameter learning

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and dynamic parameterization compared to SAC-SMA's site-by-site calibration. Data from all training basins constrain the parameterization network, improving generalization and taking advantage of "data synergy" (Fang et al., 2022)—the ability to integrate information across diverse basins to enhance predictive skill beyond what individual site-specific calibrations can achieve. Consequently, differentiable models provide a promising foundation for national-scale water modeling (Johnson et al., 2023; Song, Bindas, et al., 2025), where robust, spatially-transferable parameterizations are essential for capturing hydrologic variability across regions. However, scaling issues caused by heterogeneity within large basins, routing of streamflow along the river network, and representations of regional-scale groundwater transfer remain to be addressed to enable operational modeling (Gentine et al., 2012; Song, Bindas, et al., 2025; Song, Shen, et al., 2024).

In our unseen extreme event evaluation, δHBV trained with the KGE-based loss function had lower peak errors than those using the MNSE or RMSE loss functions, consistent with Mizukami et al. (2019). However, the KGE loss model performed worse in the general temporal test (Appendix C2), where no peak events were held out during training (Figure C2a). This suggests that the MNSE loss function is more suitable when the training data are more complete and have greater variability. Overall, δHBV1.1p with the MNSE loss reached a decent compromise between different flow magnitudes and testing scenarios. Future work can use more advanced methods to adaptively select a better loss function or to combine multiple objectives when confronted with different testing scenarios.

Overall, we found no sufficient difference in performance to select between δHBV1.1p with three (γ, β, and θ_{K_0}) versus four (γ, β, θ_{K_0}, and θ_{UZL}) dynamic parameters: both models provided similarly strong performance for the prediction of extremes in both temporal and spatial tests without significantly sacrificing accuracy in other flow magnitudes (Table 2) or untrained variables (Table 3). The use of the dynamic recession coefficient (θ_{K_0}) and maximum soil storage (θ_{UZL}) relaxed the model constraints on fast flow prediction to compensate for the simple structure of the runoff generation module. Adding one more dynamic parameter, θ_{UZL}, offered minimal benefit to peak prediction compared with adding only θ_{K_0}, and only slightly decreased the accuracy of E_T and SWE predictions. The peak prediction benefits are observed mainly in water years with return periods of less than 50 years, rather than those exceeding 50 years, suggesting that this extra dynamic parameter (θ_{UZL}) may capture some data patterns in more frequent events that are not observed in the rarer events. Such pattern shifts may be due to variations in the generation mechanisms of flows with different magnitudes across different climate regions, including short rain floods, long rain floods, excess rainfall floods, rain/snowmelt floods, snowmelt floods, and others (Stein et al., 2020). It is possible that the most extreme events were combinations of multiple mechanisms, leading to quite different runoff patterns. Due to the marginal benefits from dynamic θ_{UZL}, from a parsimonious point of view, perhaps the three-dynamic-parameter model variant is slightly more favorable. We remain cautious about adding too many dynamic parameters and would instead prefer to upgrade the model formulation if the same performance can be accomplished. However, the improved results suggest that if more systematic evaluations (spatial test, unseen cases, untrained variables) are put in place as guidance, it may be warranted to assess the use of additional dynamic parameters to see whether performance can be gained for practical prediction tasks.

Although δHBV1.1p demonstrated a state-of-the-art performance on standard hydrological data sets with small basins (catchment area <2000 km^2), its reliability in very large downstream basins as well as smaller basins remains unknown. This uncertainty arises from the use of basin-averaged inputs for training the lumped hydrological model and the significant heterogeneity of large downstream basins. Alternatively, distributed or subbasin-scale modeling, which divides the watershed into higher-resolution units and accounts for spatial variability, may be required for good performance in these basins (Song, Bindas, et al., 2025). Additionally, the current model does not include reservoir modules, with their impact temporarily compensated by the internal routing function and physical parameters such as recession coefficients and storage. In the future, it will be necessary to introduce high-resolution inputs to improve model accuracy, as well as to explicitly represent the functions of reservoirs in the routing modules. While the findings suggest that hybrid differentiable models can match or generalize better than purely data-driven models, differentiable models are hindered by their development time. Even though (with the help of NNs) their implementation may evolve faster than traditional approaches without learnable components, they nonetheless require more hypothesis testing and evaluation time than generically applicable neural networks. For example, it took a few iterations to improve δHBV beyond the initial work first reported in 2022 (Feng et al., 2022). The performance may be even further improved by

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implementing implicit numerical methods such as those allowed by an adjoint scheme (Song, Knoben, et al., 2024), albeit with more time required for testing. This suggests that there is a high potential ceiling for the approach, and that the interpretability, knowledge discovery potential, and generalizability offered by differentiable models can be earned with some extra time and effort.

In evaluating regional models, our focus is on statistical improvements in model performance. This improvement of a hybrid differentiable model over a single LSTM, although not universal, is still significant, especially considering that $LSTM_s$ surpasses one implementation of the traditional SAC-SMA model (Frame et al., 2022) by 13% when predicting rare extreme events (100+ year return period events in Figure C3 in Appendix C3), and δHBV1.1p achieved a further 10% improvement beyond $LSTM_s$. The progress made with the δHBV variants presented in this work is comparable to the progress made by $LSTM_s$ relative to SAC-SMA. Knoben et al. (2025) demonstrated that much fewer models are needed to achieve performant simulations across a large domain than earlier research may have suggested. Therefore, future work aimed at modeling large geographical regions can employ ensembles of a few differentiable models, or select better-performing single models from an array of options, including neural networks and physical or hybrid models.

5. Conclusions

In this work, we mainly tested the differentiable (hybrid) hydrologic model, δHBV1.1p, with different numbers of dynamic parameters and types of loss functions for the prediction of unseen extreme events. Overall, all versions of δHBV1.1p performed better than a purely data-driven model, $LSTM_s$, when predicting unseen events, particularly those with return periods greater than 50 years. δHBV1.1p with three dynamic parameters reduced peak flow simulation errors in 80% of the unseen rare events compared to $LSTM_s$. We argue that the improved generalization compared to the previous state-of-the-art LSTM is due to the use of physical equations like mass balances and first-order fluxes which give the model some physical realism in producing (nearly linear) responses to extreme events, enabling the model to output values that are outside the range of the training data. From another interpretation, the process equations constrain the hybrid model to respond within the physically consistent response space that can generate such responses. Furthermore, the loss function seems to have a larger impact on δHBV than LSTM, which has not previously been discussed in detail. The choice or combination of loss functions can lead to differentiable models that outperform LSTM in certain ways, but it is currently unclear whether certain loss functions are more optimal than others. We encourage future work with differentiable models to pay attention to tuning loss functions.

δHBV1.1p with three parameters (γ, β, and θ_{K_0}) provided superior performance for capturing very rare extreme events and untrained variables. Adding one more dynamic parameter, θ_{UZL}, marginally improved the model's spatial generalizability but slightly decreased the prediction accuracy of untrained variables. Contrary to concerns about significant overfitting due to the increased number of dynamic parameters, the additional parameters (θ_{K_0} and θ_{UZL}) enhanced the model's spatial generalization, which suggests that certain missing physical processes are compensated by these dynamic parameters. However, it is still unclear what the missing physical processes are, exactly, and how a general perceptual model can be built for unrepresented hydrologic behaviors. The decrease in the accuracy of untrained variables when adding more dynamic parameters also indicates that further improvements are needed in constructing the priors in differentiable models.

Even though differentiable models perform well on standard data sets with small basins (e.g., CAMELS), they also have limitations, such as an inability to account for heterogeneity in large basins and a lack of reservoir representation. Currently, no single hydrological model is likely to be optimal in all scenarios. Both pure AI models and differentiable models have their own strengths and weaknesses, which merit continued analyses; the co-evolution of AI and physical models is crucial for the advancement of both fields.

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Appendix A: Data Details

Table A1 contains the meteorological forcings and basin attributes used in all δHBV models.

Table A1
Summary of the Forcing and Attribute Variables From the CAMELS Data Set Used in All δHBV Models ($A_{attr,p}$)

Type	Variable	Description [units]
Forcings	P	Precipitation [mm/day]
	E_p	Potential evapotranspiration [mm/day]
	T	Temperature [°C]
Attributes	p_mean	Mean daily precipitation [mm/day]
	pet_mean	Mean daily E_p [mm/day]
	p_seasonality	Seasonality and timing of precipitation [-]
	frac_snow	Fraction of precipitation falling as snow [-]
	aridity	E_P/P [-]
	high_prec_freq	Frequency of high precipitation days [days/year]
	high_prec_dur	Average duration of high precipitation events [days]
	low_prec_freq	Frequency of dry days [days/year]
	low_prec_dur	Average duration of dry periods [days]
	elev_mean	Catchment mean elevation [m]
	slope_mean	Catchment mean slope [m/km]
	area_gages2	Catchment area (GAGESII estimate) [km^2]
	frac_forest	Forest fraction [-]
	lai_max	Maximum monthly mean of the leaf area index [-]
	lai_diff	Difference between the maximum and minimum monthly mean of the leaf area index [-]
	gvf_max	Maximum monthly mean of the green vegetation [-]
	gvf_diff	Difference between the maximum and minimum monthly mean of the green vegetation fraction [-]
	dom_land_cover_frac	Fraction of the catchment area associated with the dominant land cover [-]
	dom_land_cover	Dominant land cover type [-]
	root_depth_50	Root depth at 50th percentile, extracted from a root depth distribution based on the International Geosphere-Biosphere Programme (IGBP) land cover [m]
	soil_depth_pelletier	Depth to bedrock [-]
	soil_depth_statgso	Soil depth [m]
	soil_porosity	Volumetric soil porosity [-]
	soil_conductivity	Saturated hydraulic conductivity [cm/hr]
	max_water_content	Maximum water content [m]
	sand_frac	Sand fraction [-]
	silt_frac	Silt fraction [-]
	clay_frac	Clay fraction [-]
	geol_class_1st	Most common geologic class in the catchment basin [-]
	geol_class_1st_frac	Fraction of the catchment area associated with its most common geologic class [-]
	geol_class_2nd	Second most common geologic class in the catchment basin [-]
	geol_class_2nd_frac	Fraction of the catchment area associated with its 2nd most common geologic class [-]
	carbonate_rocks_frac	Fraction of the catchment area as carbonate sedimentary rocks [-]
	geol_porosity	Subsurface porosity [-]
	geol_permeability	Subsurface permeability [m^2]

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Appendix B: Method Details

B1. Long Short-Term Memory

$LSTM_s$, the LSTM for streamflow prediction from Kratzert et al. (2019) can be summarized as:

Input gate: $$i^t = \left(W_{ix}x^t + W_{ih}h^{t-1} + b_i\right) \tag{B1.1}$$

Forget gate: $$f^t = \left(W_{fx}x^t + W_{fh}h^{t-1} + b_f\right) \tag{B1.2}$$

Input node: $$g^t = \tanh\left(W_{gx}x^t + W_{gh}h^{t-1} + b_g\right) \tag{B1.3}$$

Output gate: $$o^t = \left(W_{ox}x^t + W_{oh}h^{t-1} + b_o\right) \tag{B1.4}$$

Cell state: $$s^t = g^t \odot i^t + s^{t-1} \odot f^t \tag{B1.5}$$

Hidden state: $$h^t = \tanh\left(s^t\right) \odot o^t \tag{B1.6}$$

Output: $$y^t = W_{hy}h^t + b_y \tag{B1.7}$$

where the input, x^t, is a concatenation of meteorological forcings and static attributes at time step t. Variables i, f, and o are the input, forget, and output gates of $LSTM_s$, respectively. g is the input node. s and h represent the hidden states and cell states, which carry short-term and long-term memory, respectively. tanh and σ represent the hyperbolic tangent (varying between -1 and 1) and sigmoid (varying between 0 and 1) activation functions, respectively. W and b are the weights of neurons and bias of each layer, respectively. $\odot$ is element-wise multiplication. y is the predicted variable.

$LSTM_p$, the LSTM used for parameterization in δHBV, has a similar structure to $LSTM_s$ (Feng et al., 2020, 2022). It has an additional linear layer to process the input before the input gate, and a dropout operator applied to the hidden layers to avoid overfitting. Both LSTM structures have previously been trained for streamflow prediction and showed comparable performance on the CAMELS data set (Feng et al., 2020, 2022; Kratzert et al., 2019).

Input transformation: $$x^t = \mathrm{ReLU}\left(W_I I^t + b_I\right) \tag{B1.8}$$

Input gate: $$i^t = \left(\mathcal{D}\left(W_{ix}x^t\right) + \mathcal{D}\left(W_{ih}h^{t-1}\right) + b_i\right) \tag{B1.9}$$

Forget gate: $$f^t = \left(\mathcal{D}\left(W_{fx}x^t\right) + \mathcal{D}\left(W_{fh}h^{t-1}\right) + b_f\right) \tag{B1.10}$$

Input node: $$g^t = \tanh\left(\mathcal{D}\left(W_{gx}x^t\right) + \mathcal{D}\left(W_{gh}h^{t-1}\right) + b_g\right) \tag{B1.11}$$

Output gate: $$o^t = \left(\mathcal{D}\left(W_{ox}x^t\right) + \mathcal{D}\left(W_{oh}h^{t-1}\right) + b_o\right) \tag{B1.12}$$

Cell state: $$s^t = g^t \odot i^t + s^{t-1} \odot f^t \tag{B1.13}$$

Hidden state: $$h^t = \tanh\left(s^t\right) \odot o^t \tag{B1.14}$$

Output: $$y^t = \left(W_{hy}h^t + b_y\right) \tag{B1.15}$$

I^t is the concatenation of meteorological forcings and static attributes, similar to the input, x^t, in Equation B1.1. $\mathcal{D}$ is the dropout operator, which randomly sets the weights of some neurons to zero to prevent overfitting. ReLU represents the Rectified Linear Unit (varying between 0 and $+\infty$).

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Table B1
Model Structures and Parameters of Hydrologiska Byråns Vattenbalansavdelning

Modules	Equations	Fluxes	Parameters
Snow accumulation and melt	$\frac{dS_p}{dt} = P_s + R_{fz} - S_{melt}$ $P_s = P$ if $T < \theta_{TT}$, otherwise 0 $R_{fz} = (\theta_{TT} - T)\theta_{DD}\theta_{rfz}$ $s_{melt} = (T - \theta_{TT})\,\theta_{DD}$ $\frac{dS_{liq}}{dt} = s_{melt} - R_{fz} - I_{snow}$ $I_{snow} = s_{liq} - \theta_{CWH}S_p$	S_p: storage in snow P_s: precipitation as snow R_{fz}: refreezing of liquid snow S_{melt}: snowmelt as water equivalent S_{liq}: liquid water content in the snowpack I_{snow}: snowmelt infiltration to soil moisture	θ_{TT}: threshold temperature for snowfall [°C], range: [-2.5,2.5]. θ_{DD}: degree-day factor [mm°C^{-1} day^{-1}], range: [0,0.1]. θ_{rfz}: refreezing coefficient [-], range: [0.5,10]. θ_{CWH}: water holding capacity as a fraction of the current snowpack [-], range: [0,0.2].
Soil moisture and evapotranspiration	$\frac{dS_S}{dt} = I_{snow} + P_r - P_{eff} - E_x - E_T(+C_r)$ $P_r = P$ if $T > \theta_{TT}$ otherwise 0 $P_{eff} = \min\left(\left(\frac{S_S}{\theta_{FC}}\right)^{\beta}, 1\right)(P_r + I_{snow})$ $E_x = \frac{(S_S - \theta_{FC})}{dt}$ $E_T = \min\left(\left(\frac{S_S}{\theta_{FC}\theta_{LP}}\right)^{\gamma}, 1\right)E_P$ $C_r = \theta_C * S_{LZ} * \left(1 - \frac{S_S}{\theta_{FC}}\right)$	S_S : storage in soil moisture S_{LZ}: current storage in the lower subsurface zone P_r: precipitation as rain P_{eff}: effective flow to the upper subsurface zone E_x: rainfall excess E_T: actual evapotranspiration. C_r: an upward flow from the lower subsurface zone. This flux is only used in δHBV1.1p	θ_C: time parameter [day^{-1}], range: [0,1]. θ_{FC}: maximum soil moisture (field capacity) [mm], range: [50,1000]. θ_{LP}: vegetation wilting point [-], range: [0.2,1]. β: a parameter influencing the shape of the soil moisture function [-], range: [1,6]. γ: a parameter influencing the shape of the evapotranspiration function [-], range: [0.3,5].
Runoff generation	$\frac{dS_{UZ}}{dt} = P_{eff} + E_x - perc - Q_0 - Q_1$ $perc = \min(\theta_{perc}, \frac{S_{UZ}}{dt})$ $Q_0 = \theta_{K_0}(S_{UZ} - \theta_{UZL})$ $Q_1 = \theta_{K_1}S_{UZ}$ $\frac{dS_{LZ}}{dt} = \text{perc} - Q_2(-C_r)$ $Q_2 = \theta_{K_0}S_{LZ}$ $Q' = Q_0 + Q_1 + Q_2$ $Q_b(t) = \int_0^t \xi(s)\,Q'(t-s)\,ds$ $\xi(s) = \frac{1}{\Gamma(\theta_a)\theta_b^{\theta_a}} s^{\theta_a - 1} e^{-\frac{1}{\theta_b}}$	S_{UZ}: storage in the upper subsurface zone $perc$: percolation to the lower subsurface zone Q_0: near surface flow Q_1: interflow Q_2: baseflow S_{LZ}: storage in the lower subsurface zone Q_b: simulated streamflow for basin b	θ_{perc}: percolation flow rate [mm/day], range: [0,10]. θ_{K_0} (range: [0,10]), θ_{K_1} (range: [0.05,0.9]), and θ_{K_2} (range: [0.001,0.2]): recession coefficients [day^{-1}] θ_a (range: [0,2.9]) and θ_b (range: [0,6.5]): two routing parameters [-] θ_{UZL}: maximum storage of the upper soil layer [mm], range: [0,100].

B2. Differentiable Hydrologiska Byråns Vattenbalansavdelning

Table B1 provides the details of HBV, including its equations, fluxes, and parameters. The structural difference between δHBV1.0 and δHBV1.1p is the incorporation of the upward flow, C_r, in the soil moisture and evapotranspiration module and runoff generation module.

Appendix C: Additional Results

C1. Model Evaluation With Multiple Metrics

Figure C1 shows box plots of the Nash-Sutcliffe model efficiency coefficient (NSE) and Kling-Gupta model efficiency coefficient (KGE) values for high- and low-probability water years in the temporal extrapolation and spatial tests. In general, δHBVs presented better annual NSE values than LSTM_s in low-probability water years, and the advantage increased with the return period.

C2. Benchmarking in the General Temporal Test (Without Holding Out Extremes)

In the general temporal test, where extreme events in the training period were not held out from the training data set, the δHBV model variant trained with KGE loss showed degraded performance and no longer held advantages against the variant using MNSE loss (Figure C2, Table C1). In this case, all models were trained with all available data from 1980/10/01 to 1990/09/30 and tested from 1995/10/01 to 2005/09/30 (Figure C2a) on the 531 basins from CAMELS used in Newman et al. (2015). In terms of the top 0.1% of flows for each basin, δHBV1.1p also compared favorably against LSTM_s and the results of an encoder-decoder LSTM reported in Ou et al. (2025) (Figure C2). Their model simulation utilized the encoder–decoder LSTM structure from Nearing et al. (2024). Note that Ou et al. (2025) used all three types of forcings (Daymet, Maurer, and NLDAS) from the CAMELS data

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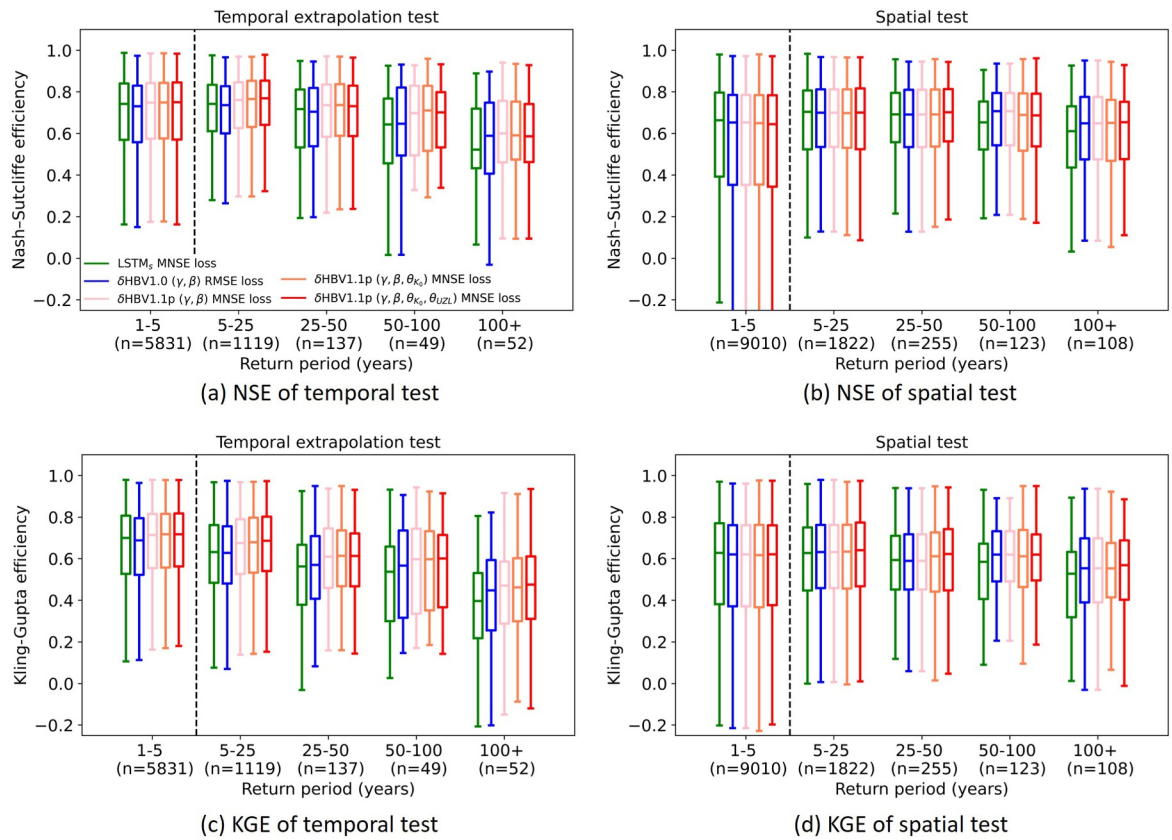

Figure C1. Box plots of annual Nash-Sutcliffe Efficiency values stratified by the return period of the water years in (a) temporal extrapolation test in the testing period of 1980/10/01–1995/09/30 and (b) spatial test on the untrained basins in the time period of 1990/10/01–2014/09/30. (c, d) are the Kling-Gupta Efficiency values for these two tests. Each box's horizontal line represents the median, while its top and bottom mark the first and third quartiles. The whiskers extend to the lowest and highest points within 1.5 times the interquartile range.

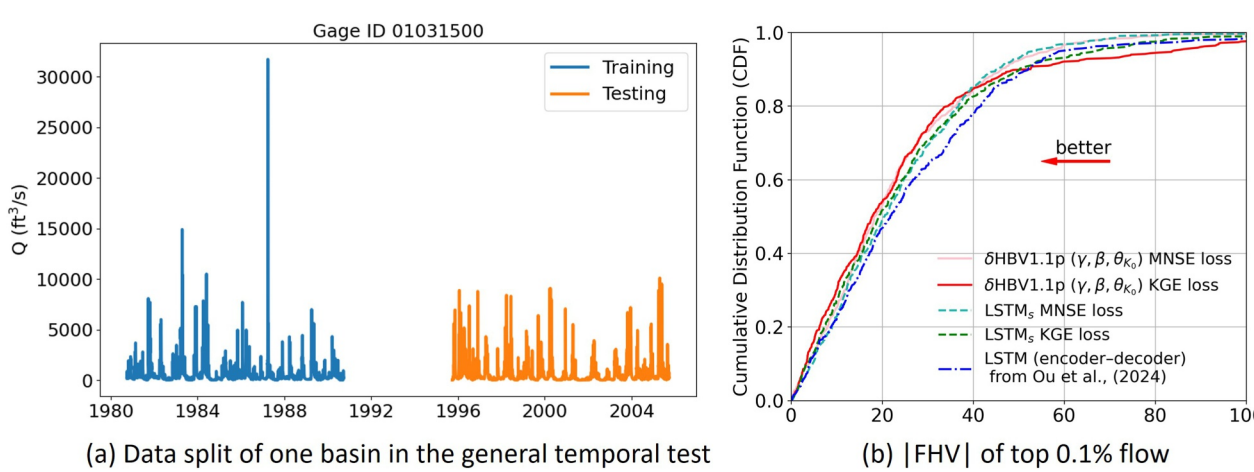

Figure C2. (a) Illustration of the data split in one basin for the general temporal test without holding out extreme data from the training data set. (b) Cumulative Distribution Function of the absolute percent bias of the top 0.1% of flows (|FHV| defined in Table 1). δHBV1.1p($\gamma, \beta, \theta_{K_0}$) and LSTM$_s$ used the Daymet forcing data from CAMELS while the result of the long short-term memory (LSTM) (encoder-decoder) streamflow model from Ou et al. (2025) used all three types of forcings (Daymet, Maurer, and NLDAS) from CAMELS. Note that the results of the LSTM (encoder-decoder) were digitized from Ou et al. (2025), and the LSTM (encoder-decoder) is originally from Nearing et al. (2024).

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Table C1
Median Daily Streamflow Metrics in the General Temporal Test Across 531 CAMELS Basins

Model	Loss function	NSE	KGE	Low flow (bottom 30%) RMSE (mm/day)	High flow (top 2%) RMSE (mm/day)
δHBV1.1p($\gamma, \beta, \theta_{K_0}$)	MNSE loss	0.739	**0.763**	0.080	2.981
δHBV1.1p($\gamma, \beta, \theta_{K_0}$)	KGE loss	0.663	0.725	0.066	3.170
$LSTM_s$	MNSE loss	**0.745**	0.759	**0.064**	3.145
$LSTM_s$	KGE loss	0.705	0.737	0.094	**2.873**

Note. All models were trained with all available data from 1980/10/01 to 1990/09/30 and tested from 1995/10/01 to 2005/09/30. Bold text indicates the best result for each category.

set, while the models in this work (δHBV1.1p and $LSTM_s$) used only Daymet. The differentiable model (δHBV1.1p) with MNSE loss provided the best performance in terms of the absolute percent bias of the top 0.1% of flows (|FHV| defined in Table 1) among all models (Figure C2). In contrast, δHBV1.1p with KGE loss increased the portion of simulated flows with large bias.

C3. Benchmarking in the Temporal Interpolation Test

In Frame et al. (2022), the temporal test (called the temporal interpolation test here) was designed to predict the holdout data within the training period, as shown in Figure C3a, rather than in a separate time span, as in the temporal extrapolation test in the main text (Figure 2a). We first reproduced the result of $LSTM_s$ as shown in Frame et al. (2022) to demonstrate we correctly used the LSTM model from Kratzert et al. (2019), and then ran the differentiable models in the temporal interpolation test (Figure C3b) to benchmark them against the Sacramento Soil Moisture Accounting (SAC-SMA) model from Frame et al. (2022). In this test, the model was trained with high-probability water years (return period <5 years) from 1990/10/01 to 2014/09/30 and tested on the low-probability water years (return period >5 years) held out of the training data set from 1996/10/01 to 2014/09/30. Since the holdout data were interspersed among the training data (Figure C3a), the LSTM model gained some benefits by remembering the data patterns and noise in the training data, slightly outperforming δHBV1.1p when the return period was less than 25 years. However, δHBVs still surpassed it in rarer scenarios (return period

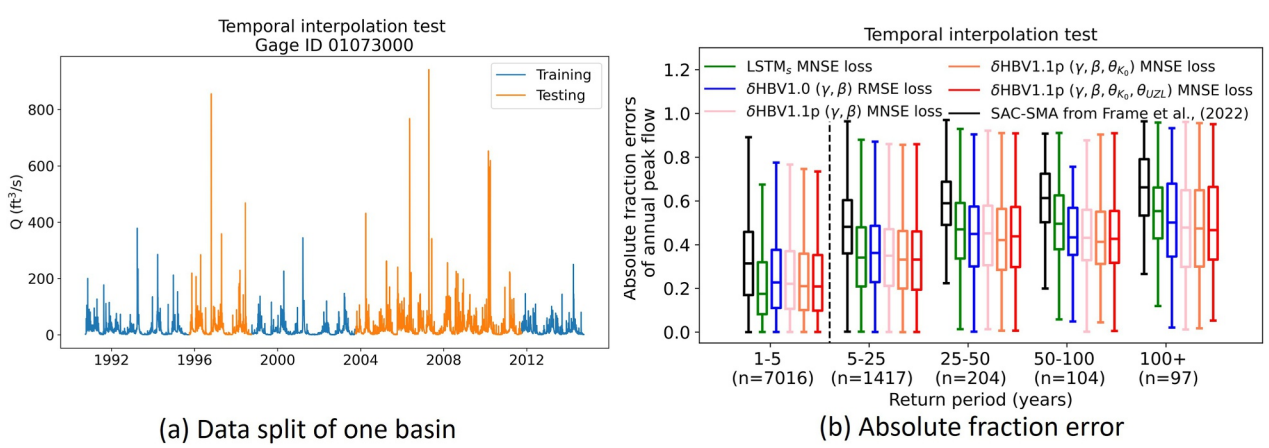

Figure C3. (a) Illustration of the data split for the temporal test in Frame et al. (2022), called the temporal interpolation test in this work, for clarity. (b) Box plots of the absolute fraction errors of annual peak flow stratified by the return period of the water years in the temporal interpolation test in Frame et al. (2022). The model was only trained in the water years within the training time period that also had return periods of less than 5 years. The left-most panel shows the peak errors calculated on the training data, while the panels to the right are the peak errors for the water years (n) with 5+ year return periods in the training time. The long short-term memory boxplots reproduce what Frame et al. (2022) showed in their paper corrigendum (Frame et al., 2023), while the SAC-SMA boxplots were digitized from their paper. Each box's horizontal line represents the median, while its top and bottom mark the first and third quartiles. The whiskers extend to the lowest and highest points within 1.5 times the interquartile range.

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>25 years). $LSTM_s$ surpassed the traditional SAC-SMA model by 13% in the low-probability events (return period >100 years), while δHBV1.1p achieved a further 10% improvement over $LSTM_s$.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Data Availability Statement

The code for the δHBV models is available at https://doi.org/10.5281/zenodo.15978036 (Song, Feng, & Shen, 2025). The main code is in dPLHBVrelease-master\hydroDL-dev\example\dHBV1.1p.ipynb. The code for the LSTM model can be downloaded at https://doi.org/10.5281/zenodo.6326394. The code for return period calculation and for training/testing data splitting is from Frame et al. (2023) and is available at https://doi.org/10.5281/zenodo.7314083. The CAMELS data set can be downloaded at https://dx.doi.org/10.5065/D6MW2F4D (Addor et al., 2017; Newman & Clark, 2014). MODIS ET data can be downloaded at https://modis.gsfc.nasa.gov/data/dataprod/mod16.php (Running et al., 2017). The gridded SWE data can be downloaded at https://nsidc.org/data/nsidc-0719/versions/1 (Broxton et al., 2016).

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